How to install Terraform?

sudo apt update

sudo apt install -y gnupg software-properties-common curl

Add HashiCorp GPG key (NEW METHOD)

curl -fsSL https://apt.releases.hashicorp.com/gpg | \

sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg

Add Terraform repository

echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] \
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | \
sudo tee /etc/apt/sources.list.d/hashicorp.list

sudo apt update

sudo apt install terraform

terraform -version

nano main.tf

```
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "terraform_ec2" {
  ami           = "ami-053b0d53c279acc90"
  instance_type = "t2.micro"

  tags = {
    Name = "Terraform-Demo-Server"
  }
}
```

Ls – don’t work

Ls -a

.terraform
.terraform.lock.hcl
main.tf

Purpose:

.terraform/

stores downloaded provider plugins.

terraform validate
terraform plan
terraform apply

Step 2 — Preview Infrastructure

Run:

terraform plan

Terraform will read your code:

Will read this line ---→ resource "aws_instance" "terraform_ec2"

In planning u might get error on authentication

Install azure cli in system.

https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html

Now verify the installation of Aws cli -> awscli –version

Configure access key and secrate key

Aws configure

https://k21academy.com/aws-cloud/create-access-and-secret-keys-in-aws/

run

terraform plan

terraform apply

aws ec2 describe-instances

aws ec2 describe-instances --query 'Reservations[*].Instances[*].[InstanceId,ClientToken]' --output table

give me specific instance details - aws ec2 describe-instances --instance-ids i-0f6c6604e0ee6abd4

now Lets check terraform state –

go inside the project folder.

terraform state list

Terraform + AWS EC2 Revision Notes

1. What is Terraform

Terraform is an **Infrastructure as Code (IaC)** tool developed by HashiCorp used to provision and manage cloud infrastructure using declarative configuration files.

Key features:

- Infrastructure as Code
- Multi-cloud support (AWS, Azure, GCP)
- State management
- Automation of infrastructure lifecycle

Terraform Architecture

```
Terraform Configuration (.tf)
        |
        ▼
Terraform Core
        |
        ▼
Providers (AWS, Azure, GCP)
        |
        ▼
Cloud APIs
```

Components:

Component	Description
Terraform Core	Reads configuration and manages state
Providers	Plugins that interact with cloud APIs
Resources	Infrastructure objects like EC2, S3
State	Tracks real infrastructure

Terraform File Structure

Example project:

```
terraform-ec2-project/
|
 ├── main.tf
 ├── terraform.tfstate
 ├── terraform.lock.hcl
 └── .terraform/
```

Explanation:

File	Purpose
main.tf	Terraform configuration
terraform.tfstate	Tracks infrastructure
terraform.lock.hcl	Locks provider versions

File	Purpose
.terraform	Stores provider plugins

Basic Terraform EC2 Configuration

Example:

```
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "terraform_ec2" {
  ami           = "ami-053b0d53c279acc90"
  instance_type = "t2.micro"

  tags = {
    Name = "Terraform-Demo-Server"
  }
}
```

Explanation:

Block	Meaning
provider	Cloud provider configuration
aws_instance	EC2 resource
ami	OS image
instance_type	Server size
tags	Metadata for AWS resources

Terraform Workflow

Standard workflow used by DevOps engineers:

```
Write Code
    |
    ▼
terraform init
    |
    ▼
terraform validate
    |
    ▼
terraform plan
    |
    ▼
terraform apply
    |
```

▼
terraform destroy

Important Terraform Commands

Initialize Project

terraform init

Purpose:

- Download providers
- Prepare working directory

Validate Configuration

terraform validate

Checks syntax and configuration correctness.

Preview Infrastructure

terraform plan

Shows what Terraform **will create or modify**.

Example:

Plan: 1 to add, 0 to change, 0 to destroy

Create Infrastructure

terraform apply

Creates infrastructure defined in .tf files.

Destroy Infrastructure

terraform destroy

Deletes resources tracked in Terraform state.

Terraform State

State file:

terraform.tfstate

Purpose:

- Tracks infrastructure resources
- Maps configuration to real-world resources

Example state entry:

aws_instance.terraform_ec2

Useful Terraform State Commands

List Resources

terraform state list

Example:

aws_instance.terraform_ec2

Show Resource Details

terraform state show aws_instance.terraform_ec2

Shows attributes like:

- instance ID
- public IP
- instance type

AWS CLI Commands Used

Configure AWS Credentials

aws configure

Stores credentials in:

~/.aws/credentials

Check EC2 Instances

aws ec2 describe-instances

Clean Output

```
aws ec2 describe-instances \
--query 'Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress]' \
--output table
```

Authentication Methods for Terraform

Terraform can authenticate to AWS using:

1. AWS CLI credentials
2. Environment variables
3. IAM roles
4. Terraform Cloud

Example environment variables:

```
export AWS_ACCESS_KEY_ID=XXXX
export AWS_SECRET_ACCESS_KEY=XXXX
```

Best Practice (Production)

Instead of storing keys, companies attach **IAM roles to EC2 instances**.

Architecture:

```
Terraform Server (EC2)
      |
      ▼
   IAM Role
      |
      ▼
   AWS APIs
```

Benefits:

- No secret keys
- Better security
- Automatic credential rotation

Common Interview Questions

1. What is Terraform?

Terraform is an Infrastructure as Code tool used to provision and manage cloud infrastructure using declarative configuration files.

---

2. What is Terraform State?

Terraform state is a file that tracks the mapping between Terraform configuration and real infrastructure resources.

---

3. Difference Between terraform plan and terraform apply

Command	Purpose
terraform plan	Shows what changes will happen
terraform apply	Actually creates infrastructure

---

4. Why is Terraform State Important?

Without state Terraform cannot track existing resources or manage updates.

---

5. What is a Provider in Terraform?

Providers are plugins that allow Terraform to interact with external platforms such as AWS, Azure, or Kubernetes.

---

6. What Happens During terraform init?

Terraform downloads provider plugins and initializes the working directory.

---

7. What Does terraform destroy Do?

It deletes all resources recorded in the Terraform state file.

---

8. Where Are AWS Credentials Stored After aws configure?

~/.aws/credentials

Common Interview Scenario Question

Q: What happens if someone deletes an EC2 instance manually from AWS console but Terraform state still exists?

Answer:

Terraform state becomes inconsistent. Terraform may try to recreate or fail during the next run. This is called **state drift**.

Quick Revision Summary

Terraform = Infrastructure as Code
Provider = Cloud API plugin
Resource = Infrastructure object
State = Infrastructure tracking file

Workflow:

init → validate → plan → apply → destroy